Chemistry 3A – Spring 2026 **Version A**

Sections 19320-19321

Midterm Examination #4

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt.
TEAR OFF LAST PAGE and use it as scratch for writing calculations.

1. In a thermochemical equation, the energy value written on the reactant side indicates:
   a) The reaction is endothermic b) The reaction requires a catalyst
   c) The reaction is exothermic d) The reaction occurs at constant volume.

2. Which statement accurately defines the **limiting reactant** in a chemical reaction?
   a) The reactant with the largest molar mass used in the reaction
   b) The reactant remaining in the largest amount after the reaction is complete
   c) The reactant with the smallest coefficient in the balanced chemical equation
   d) The reactant that determines the maximum amount of product that can be formed

3. At a certain temperature, the reaction H_2 (g) + Cl_2 (g) ⇌ 2 HCl (g) reaches equilibrium and concentrations are determined: [H_2] = 0.0001 M, [Cl_2] = 0.0001 M, and [HCl] = 10,000 M. Compute the value of equilibrium constant *K*?
   a) 1×10^{12} b) 1×10^{-12} c) 1×10^{-16} d) 1×10^{16}

4. Which of the following is a FALSE statement about acid-base reactions?
   a) Titration is a quantitative technique that can determine the amount of moles of acid reacting with a base
   b) The products of the neutralization reaction are water (H_2O) and a salt
   c) There is a movement of a proton (H+) from the acid to the base
   d) Red litmus paper turning blue indicates a base, not an acid

5. Which of the following pairs are conjugates in acid-base chemistry? (hint: weak acid or weak base)
   a) HCl / Cl^- b) NH_4^+ / CH_4 c) H_2SO_4 / HSO_4^- d) CH_3COOH / CH_3COO^-

6. Which statement is true about pure water (H_2O)?
   a) It has an equilibrium constant K_w whose value = 10^{-7}
   b) It has a pH = 7.0
   c) It has a [H^+] = 10^{-14} *M*. ([H+] is also [H_3O^+])
   d) It has the ability to act as an acid, but not as a base

7. For the balanced reaction : CH_4 (g) + 4 Cl_2 (g) → CCl_4 (l) + 4 HCl (g), you start with 100.0 g methane (CH_4). How much **carbon tetrachloride** (CCl_4) will you produce? The molar masses (g/mol) are given: CH_4: 16.04, Cl_2: 70.91, CCl_4: 153.81, HCl: 36.46. (Hint: you should only need two of the four)
   a) 239.7 g b) 10.43 g c) 958.9 g d) 153.8 g

8. Which of the following statements is TRUE about atomic fission and atomic fusion
   a) The sun generates energy through atomic fission reactions
   b) Nuclear power plants generate energy through atomic fission reactions
   c) Nuclear power plants generate energy through atomic fusion reactions
   d) The use of enriched uranium-235 isotope is a good example of a fusion chain reaction

9. Of the statements below, select the best that is true of $^{31}_{15}P$?
   a) It has 46 neutrons and protons
   b) It has 31 protons
   c) It has 16 neutrons
   d) It has 15 neutrons

10. A solution has $[H^+] = 1.0 \times 10^{-9}$ M. What is the pH?
   a) 1.00 b) 5.00 c) 9.00 d) 14.00

11. Which of the following is a strong acid?
   a) KOH b) HCl c) NaCl d) CO_2

12. 100.00 mL of acetic acid solution was titrated with 0.100 M NaOH solution with the initial buret reading at 3.00 mL and the final buret reading at 18.00 mL. What is the acetic acid concentration?
   a) 0.00150 M b) 0.0150 M c) 0.0300 M d) 0.150 M

13. If a weak acid has an equilibrium constant $K_a = 5.0 \times 10^{-8}$, then what is the equilibrium constant K_b of its conjugate base? Note $K_w = 1.0 \times 10^{-14}$
   a) 2.0×10^{-6} b) 2.0×10^{-7} c) 5.0×10^{-7} d) 5.0×10^{-22}

14. Consider the following equilibrium: PCl_3 (g) + Cl_2 (g) ⇌ PCl_5 (g). If large amount of PCl_3 reactant is added to container, in which direction will the reaction shift to restore equilibrium?
   a) Shift to the right (forward reaction) to produce more PCl_5.
   b) Shift to the left (reverse reaction) to consume the added PCl_3.
   c) There will be no shift, as adding PCl_2 reactant does not affect the equilibrium
   d) The equilibrium constant *K* will decrease

15. Which type of radiation has the most ***ionizing*** power?
   a) Alpha particles
   b) Beta particles
   c) Gamma rays
   d) None of the above

16. What is the radioisotope of an element that is a gas that can accumulate in homes and is responsible particularly for causing nonsmoking-related lung cancers?
   a) Xenon-133 b) Krypton-85 c) Radon-222 d) Carbon-14

17. X, Y, and Z are gaseous compounds in the reaction X (g) + Y (g) ⇌ **2** Z (g). What will happen if the pressure is **increased** on the system according to Le Chatelier's Principle?
   a) The reaction will favor the formation of the reactants (the reverse direction)
   b) The reaction will favor the formation of the products (the forward direction)
   c) The reaction will change the reaction to **2** X (g) + Y (g) ⇌ Z (g)
   d) There will be no effect: the amounts of reactants and product remain the same

18. Initially you have 800.0 g of U-235 radioisotope. How much do you have left if four (4) half-lives pass?
   a) 800.0 g b) 400.0 g c) 100.0 g d) 50.00 g

19. One of the statements below is FALSE about what a buffer is and its properties?
  a) The best buffers are weak acids or bases their conjugate pairs
  b) it can resist changes in solution pH caused by addition of strong acids or bases
  c) It is a term that applies to shielding for all radiation, alpha and beta particles and gamma rays
  d) Formic acid and its conjugate base formate were mentioned as examples of buffers

20. For the equilibrium $Ca(OH)_2$ (s) ⇌ Ca^{2+} (aq) + 2 OH^- (aq), K_{sp} = 5.0 x 10^{-6}. (The units are not given usually, but value is K_{sp} = 5.0 x 10-6 M^3.) If $[Ca^{2+}]$ = 1.0 × 10^{-1} M, what is $[OH^-]$?
  a) $1.0 \times 10^{-1}\,M$ b) $5.2 \times 10^{-5}\,M$ c) $7.1 \times 10^{-3}\,M$ d) $2.5 \times 10^{-5}\,M$

21. Which one of the choices below is a TRUE statement about a catalyst?
  a) It *increases* the activation energy (E_a) seen in the reaction coordinate diagram
  b) it is *never* consumed or produced in the chemical reaction like reactants or products
  c) It *stops* a chemical reaction immediately
  d) it affects the enthalpy (ΔH) of the reaction

22. What is TRUE about the term ***molarity*** and its use?
  a) The units of molarity are "mol/L" (moles per liter)
  b) it is used to indicate the concentration of a solution
  c) if you multiply it by a volume in liters, you can get the amount (moles) of a substance
  d) all of the above

23. What is FALSE about the concept of chemical equilibrium?
  a) it indicates chemical reactions can go in a forward and reverse direction
  b) when equilibrium is attained, the rate of the forward and reverse reactions are equal
  c) it only applies to what happens when the reactants and products are gases
  d) equilibrium can be quantitatively understood by a constant *K*

24. According to the Bronsted-Lowry definition of acids and bases, which statement is TRUE?
  a) an acid is a proton acceptor
  b) an acid is a proton donor
  c) a base is a proton donor
  d) none of the above

25. Which of the following factors is LEAST LIKELY IF AT ALL to affect the rate of a chemical reaction?
  a) temperature
  b) concentration
  c) orientation of collisions
  d) the geometry of the container where the reaction is done

Continue to next page for True-False questions

26. A catalyst will reduce the enthalpy ΔH of either an endothermic or exothermic reaction
    a) true b) false

27. If the temperature of an endothermic reaction at equilibrium is increased, the equilibrium will shift in the endothermic direction.
    a) true b) false

28. **Gamma rays** are the most penetrating type of common radiation and require thick concrete or lead shielding
    a) true b) false

29. Strong acids and like HCl and strong bases like NaOH are called that because in aqueous solutions, they exist completely as ions and not as HCl (aq) or NaOH (aq)
    a) true b) false

30. Balancing a chemical reaction equation is done by a process of adjusting the coefficients of all of the compounds in the chemical reaction equation.
    a) true b) false

31. The half-life of a radioisotope is the time required for half of the sample to decay.
    a) true b) false

32. In performing a chemical reaction, the percent yield is the theoretical yield divided by the actual yield times 100 percent (100%)
    a) true b) false

33. Collision theory in part states that for a chemical reaction to occur, particles must collide with correct orientation and sufficient energy
    a) true b) false

34. Water (H_2O) can act as both an acid and a base depending on the reaction partner
    a) true b) false

35. The sum of the pH and the pOH is always 14.
    a) true b) false